

|                                                                      |                                                                |
|----------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Program: Diploma in Integrated Circuit Design and Fabrication</b> |                                                                |
| Course Code: <b>5378</b>                                             | Course Title: <b>Microcontroller &amp; Embedded System Lab</b> |
| Semester: <b>5</b>                                                   | Credits: <b>1.5</b>                                            |
| Course Category: <b>Program Core</b>                                 |                                                                |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>                           | Periods per semester: <b>45</b>                                |

### Course Objectives:

1. To equip the student with the concept of coding microcontrollers.
2. To develop the ability to design microcomputer-based embedded systems. This class allows students to learn microcomputer interfacing from both hardware and software perspective.

### Course Prerequisites:

| Topic                                                      | Course Code | Course Title                     | Semester |
|------------------------------------------------------------|-------------|----------------------------------|----------|
| Concept of microcontroller programming                     | 2039        | Electronics Tinkering workshop   | 2        |
| Number system and codes, combinational sequential circuits | 3044        | Digital Electronics theory & Lab | 3        |
| Basic Programming                                          | 3379        | Python Programming Lab           | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn | Description                                                                      | Duration (Hours) | Cognitive level |
|-----|----------------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Demonstrate usage of IDE compatible with 8051 for embedded coding and debugging. | 9                | Applying        |
| CO2 | Make use of AVR studio to exercise coding and debugging of AVR microcontroller   | 7.5              | Applying        |

|     |                                                                                              |      |          |
|-----|----------------------------------------------------------------------------------------------|------|----------|
| CO3 | Develop embedded C programs to interface various sensors and motors with AVR microcontroller | 9    | Applying |
| CO4 | Realize interfacing of display devices, keyboard and ADC with AVR microcontroller            | 13.5 | Applying |
|     | Lab Exam                                                                                     | 6    |          |

### CO - PO Mapping:

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     | 3   |     |     |     |
| CO2             | 3   |     | 2   | 3   |     |     |     |
| CO3             | 3   |     |     | 3   |     |     |     |
| CO4             | 3   |     |     | 3   |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline:

| Module Outcomes | Description                                                                             | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>Demonstrate usage of IDE compactible with 8051 for embedded coding and debugging</b> |                  |                 |
| M1.01           | Demonstrate installation, configuring, and usage of Keil $\mu$ Vision.                  | 3                | Understanding   |
| M1.02           | Make use of basic programming concept to perform arithmetic, logic operations.          | 3                | Understanding   |
| M1.03           | Make use of basic programming concept to perform code conversions.                      | 3                | Understanding   |
| CO2             | <b>Make use of AVR studio to exercise coding and debugging of AVR microcontroller</b>   |                  |                 |
| M2.01           | Familiarize AVR studio for AVR programming                                              | 1.5              | Understanding   |
| M2.02           | Develop programs for LED blinking                                                       | 1.5              | Applying        |

|            |                                                                                                     |     |               |
|------------|-----------------------------------------------------------------------------------------------------|-----|---------------|
| M2.03      | Read the status of a switch and display it on an LED                                                | 1.5 | Applying      |
| M2.04      | Develop programs for performing code conversions and data serialization                             | 3   | Applying      |
|            | Lab Exam 1                                                                                          | 3   |               |
| <b>CO3</b> | <b>Develop embedded C programs to interface various sensors and motors with AVR microcontroller</b> |     |               |
| M3.01      | Realize Temperature sensor interfacing with AVR                                                     | 1.5 | Applying      |
| M3.02      | Demonstrate interfacing of Relay with AVR                                                           | 1.5 | Applying      |
| M3.03      | Demonstrate DC Motor interface with AVR                                                             | 1.5 | Applying      |
| M3.04      | Illustrate Servo Motor Interfacing with AVR                                                         | 1.5 | Applying      |
| M3.05      | Realize Stepper Motor Interfacing with AVR                                                          | 3   | Applying      |
| <b>CO4</b> | <b>Realize interfacing of display devices, keyboard and ADC with AVR microcontroller</b>            |     |               |
| M4.01      | Realize Seven segment display interface with AVR                                                    | 3   | Applying      |
| M4.02      | Demonstrate LCD interfacing with AVR                                                                | 3   | Applying      |
| M4.03      | Illustrate Matrix Keyboard interfacing with AVR                                                     | 3   | Applying      |
| M4.04      | Realize ADC interfacing with AVR                                                                    | 3   | Applying      |
| M4.05      | Familiarize SPI interfacing with AVR                                                                | 1.5 | Understanding |
|            | Lab Exam 2                                                                                          | 3   |               |

### Student Activity

#### Suggested Open-ended Experiments:

Open-ended experiments as individual or group of 2-3. There shall be no duplication in experiments. This can be evaluated for the purpose of continuous internal evaluation at a score of 15 marks. Students should document the experiment on the record book.

Develop projects like:

1. Automatic sanitizer dispenser
2. Automatic water level controller
3. Automatic glass door system
4. Temperature controlled DC fan
5. Line follower robot
6. Automated wheel chair with hand held control panel

### Text / Reference Books

| T/R | Book Title/ Author                                                                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1  | Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition |
| T2  | K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education India, 2016                                                                      |
| T3  | Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, 2003                                                                               |
| R1  | Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, 2003                                                                               |
| R2  | Michael J. Pont, Embedded C, Pearson Education, Second Edition                                                                                           |

### Online Resources:

| Sl.No | Website Link                                                                                                                                                                          |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.tutorialspoint.com">https://www.tutorialspoint.com</a>                                                                                                           |
| 2     | <a href="https://www.electronicshub.org">https://www.electronicshub.org</a>                                                                                                           |
| 3     | <a href="https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1- beginners/">https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1- beginners/</a> |
| 4     | <a href="https://www.tutorialspoint.com/embedded_systems/index.htm">https://www.tutorialspoint.com/embedded_systems/index.htm</a>                                                     |
| 5     | <a href="https://embeddedschool.in/avr-microcontroller-programming/">https://embeddedschool.in/avr-microcontroller-programming/</a>                                                   |
| 6     | <a href="https://www.electronicwings.com/avr-atmega/atmega1632-usart">https://www.electronicwings.com/avr-atmega/atmega1632-usart</a>                                                 |